



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Sixth Semester- Diploma in Engineering (Electrical Engineering) Session 2020-21

SPECIALIZATION – Electrical Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Mid Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EE-6001	INSTALLATION, MAINTENANCE AND TESTING	4	1	2	6	70	20	10	30	20	150
2	EE-6002	ENERGY CONSERVATION AND MANAGEMENT	4	-	-	4	70	20	10	-	-	100
3	EE-6003	ELECTRIC TRACTION / PROGRAMMABLE LOGIC CONTROLLER	4	1	2	6	70	20	10	30	20	150
4	EE-6004	PROJECT	-	-	20	10	-	-	-	200	100	300
6	EE-6005	PROFESSIONAL ACTIVITIES	-	-	2	-	-	-	-	-	-	-
		Total	12	02	24	26	210	60	30	260	140	700

Theory Credits: 14
Practical Credits: 12
Total Credits: 26

Theory Marks: 210
Practical Marks: 260
Quiz, Mid Term, : 230

Total: 700



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

DIPLOMA IN ELECTRICAL ENGINEERING

(Courses effective from Academic Year 2019-20)

SYLLABUS OF COURSES TO BE OFFERED “DIPLOMA IN ENGINEERING”

Disclaimer: The graduate programme in Diploma in Engineering syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

Visit us at: www.bhabhauniversity.edu.in, Mail Id info@bhabhauniversity.edu.in



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EE-6001 INSTALLATION, MAINTENANCE AND TESTING

UNIT-I

Installation - Types of heavy Electrical equipment, unloading accessories precautions for unloading, installation of small and large machines of both static and rotating type.
Installation of pole mounted transformer.

UNIT-II

Commissioning - Tests required before commissioning procedure to be adopted for commissioning the electrical equipment in respect of -

- a) Mechanical fixture and alignment.
- b) Electrical tests.
- c) Initial precautions for starting.

Trouble Shooting - Normal performance of equipment, trouble shooting internal and external faults, instruments and accessories for trouble shooting, trouble shooting charts.

UNIT-III

Earthing - Reasons of earthing, earthing system, earth lead and its size, permissible earth resistance for different installations, improvement of earth resistance, double earthing, earth resistance measurement, rules for earthing.

Insulation testing and maintenance - Instruments used for measuring insulation resistance, reasons for deterioration of insulation resistance, improving insulation resistance, drying of insulation, Measurement of internal temperature of winding, vacuum impregnation / filtering of insulating oil, testing of insulating oil.

UNIT-IV

Preventive maintenance and environmental pollution prevention- Concepts of preventive maintenance, advantages, preventive maintenance schedule for transformer, induction motor, transmission line, circuit breaker and under- ground cable. Preventive measures to control environmental pollution results due to production of smokes gases flow of waste material and automatic reactions in research stations, plants, electrical and electronic equipments and accessories.

UNIT-V

Electrical Accidents and Safety Measures - Electrical accidents, Safety regulations, treatment of shock, fire extinguishers.

Testing and maintenance of Relays and Circuit Breakers - Testing of Relays Factory test, commissioning test and preventive periodic maintenance test. Testing of circuit breakers, voltage test, type test, preventive maintenance of circuit breaker.

Hot Line Maintenance - Meaning and advantages, special types of non- conducting materials used for tools for hot line maintenance.

LIST OF EXPERIMENTS:-

1. Maintenance of Overhead Lines.
2. Maintenance of switchgear OCB
3. Maintenance of distribution transformer in distribution system.
4. Routine / Preventive maintenance of induction motor in textile mills / industrial establishments.

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- a) Shut down and energizing procedure.
- b) Accident report writing.
- c) Permit to work.
- d) Fire extinguisher.
- 5. Insulation oil testing.
- 6. Earth resistance testing.
- 7. Test report of electrical installation. Maintenance schedule.

References Books:-

- 1) Electrical Installations work by T.G. Ffancist. E.L.B.S (Vth metric edition)
- 2) Electrical Installations Maintenance & fault location work book by T.T.T.I.(W.R.) Bhopal
- 3) Preventive maintenance Electrical equipment by Charies J Hurburt.

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EE-6002 ENERGY CONSERVATION AND MANAGEMENT

UNIT-I

Energy Scenario- Various types of renewable and non-renewable energy, energy consumption and use pattern, energy consumption and environment.

Energy Management and audit-Energy Management and its objectives, energy audit, need of energy audit, types of energy audit, energy auditing instruments.

UNIT-II

Waste heat recovery-Sources of waste heat, advantages of waste heat recovery, commercial waste heat recovery devices-Recuperators, Heat regenerators, heat pumps etc. Agricultural use of waste heat.

Heating ventilation and air conditioning-Definition of Heating, ventilation and air conditioning, Energy saving opportunities in Heating ventilation and air conditioning, Conducting Audit in Heating ventilation and air conditioning.

UNIT-III

Role of maintenance in energy conservation-Types of maintenance- breakdown, predictive & preventive, maintenance and energy conservation.

Demand side management –Benefits, Demand side management Techniques, implementation of Demand side management program, Tariff options of Demand side management.

UNIT-IV

Energy efficient motor and drives-Motor efficiency, energy efficient motors, energy efficient electric drives, use of variable speed drives.

Power factor improvement-Causes of low power factor, advantages of power factor improvement, methods of power factor improvement.

UNIT-V

Energy conservation in various sectors-

- For residential and commercial sector in transportation in energy intensive industries.
- Co-Generation benefits, types of co-generation.

Economic Analysis of energy conservation-

Economic analysis of investment, Economic analysis techniques, Risk analysis.

References Books:-

- 1) Energy Conservation and Management by S. K. Soni and Manoj Nair, Satya Prakashan, New Delhi
- 2) Energy management- W.R.Murphy & G.M. ckey, Butter worths
- 3) Electrical Energy utilization & conservation – Dr. S.C.Tripathi

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EE-6003 ELECTRIC TRACTION

UNIT-I

General Description of Electric Traction system in India- Electric Traction advantage and disadvantages Choice of traction system in India. **System of Track Electrification.** Description of various systems D.C., 1-Phase low frequency A.C., 1-Phase high frequency, 3-Phase A.C. and Composite system. 25 K.V. A.C. 50 Hz System-Advantages and disadvantage Problems associated with A.C traction system, current and voltage unbalance, production of harmonics and induction effects, comparison between A.C. and D.C. system **Power Supply Arrangements** High Voltage Supply. Constituents of supply system substation, feeding post, feeding and sectioning arrangements, sectioning post, elementary section. Miscellaneous equipment at control posts and switching station. Major equipment at substation, transformer, circuit breaker, interrupters. Protection system for A.C. Traction.

UNIT-II

A.C. Electric Locomotive. Block diagram of A.C. electric locomotive, Overhead equipment (O.H.E.) Pentagonal O.H.E.- catenary construction. OHE Supporting structure Current collection system, current collection gear for OHE, pole collection bow collection, pantograph collector. Air blast C.B. Tap Changer (on load) Transformer Rectifier connection Traction motor connection Smoothing reactor Desirable characteristics of traction motors Traction motors-suitability of motors for traction, D.C. Series motors, A.C. Series single phase, repulsion motor, 3-phase I.M. linear I.M.

UNIT-III

Train signaling System of train lighting, special requirements of train lighting methods of obtaining unidirectional polarity and constant output. Battery System. Failure of under frame generating equipment

UNIT-IV

Traction Mechanics. Types of service Speed time curve Simplified speed time curve Average speed and schedule speed Tractive effort Power of traction motor Specific energy consumption, factors affecting specific energy consumption. Mechanics of train movement Coefficient of adhesion, factors affecting the coefficient of adhesion.

LIST OF EXPERIMENTS:-

- 1) Draw Speed Current Characteristic Of D.C. Series Motor.
- 2) Draw Speed Torque Characteristic Of D.C. Series Motor.
- 3) Study Of Various Methods For Speed Control Of D.C.
- 4) Study Of Pole And Bow Current Collector.
- 5) Study Of Pantograph Current Collector.
- 6) Study Of Metadyne Control System.

References Books:-

- 1) Electric Traction by A.T. DOVER, Publisher PITMIN & SONS
- Electric Traction System Equipment by D.W. HINGLE, Publisher PERGAMO PRES

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